

Inefficiency in Packing Squares with Unit Squares

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It is shown that, in packing a square of side $n + \frac{1}{2}$ with unit squares, the wasted space always has area $\gg n^{1/2}$. This answers a question of Erdős and Graham.

1. INTRODUCTION

Following Erdős and Graham [1] we define, for each real number $\alpha \geq 1$,

$$w(\alpha) = \alpha^2 - \sup_{\mathcal{A}} |\mathcal{A}|, \quad (1)$$

where $\mathcal{A}$ ranges over all packings of unit squares into a given square $S(\alpha)$ of side α and $|\mathcal{A}|$ denotes the number of unit squares in $\mathcal{A}$. They show that $w(\alpha) = O(\alpha^{7/11})$ (Montgomery has an unpublished argument which enables him to replace the $7/11 = 0.636363\dots$ by $(3 - \sqrt{3})/2 + \epsilon = 0.633974\dots + \epsilon$), but are unable to rule out the possibility that $w(\alpha) = O(1)$. They also speculate that the correct bound is $O(\alpha^{1/2})$. We are dubious as to the validity of such a small bound, but are unable to prove that it is false. However, we are able to show that if true, then it is essentially best possible.

THEOREM. *Suppose that $\alpha(\alpha - [\alpha]) > 1/6$. Then*

$$w(\alpha) \gg (\|\alpha\| \alpha)^{1/2}$$

where $\|\alpha\|$ denotes the distance of α from the nearest integer.

We remark that if $\alpha(\alpha - [\alpha]) \leq 1/6$, then

$$\alpha^2 - [\alpha]^2 \leq 2\alpha(\alpha - [\alpha]) \leq \frac{1}{3}$$

so that $\sup_{\mathcal{A}} |\mathcal{A}| = [\alpha]^2$. Thus $w(\alpha) \gg \alpha(\alpha - [\alpha])$ and this is essentially best possible. We also observe that if $\alpha(1 + [\alpha] - \alpha) \leq 1/6$, then

$$1 > \alpha^2 - ([\alpha] + 1)^2 + 1 = 1 - (\alpha + [\alpha] + 1)(1 + [\alpha] - \alpha) \geq \frac{1}{2}.$$

Hence the number of squares in any packing $\mathcal{A}$ is at most $([\alpha] + 1)^2 - 1$, and $w(\alpha) \geq \frac{1}{2} \geq (\alpha(1 + [\alpha] - \alpha))^{1/2}$. Thus we assume henceforward that

$$\|\alpha\| \alpha > 1/6. \quad (2)$$

2. NOTATION AND TERMINOLOGY

We assume that $S(\alpha)$ is a closed square of side $\alpha \geq 1$. By a packing $\mathcal{A}$ we mean an aggregate of closed squares of side 1, all contained in $S(\alpha)$, whose interiors are pairwise disjoint. We call $S(\alpha)$ the *containing square* and deem a pair of opposite sides to be horizontal. We then refer to its vertical, right, left, top and bottom sides in the obvious way. We use S_0 for the boundary of $S(\alpha)$.

By a *square* we henceforward mean a member of the packing. The *inclination* of a square is the minimum of the angles that the sides of the square make with the boundary of the containing square. Similarly the inclination between two squares is the minimum of the angles that the sides of one make with the sides of the other. Observe that in neither case can the inclination exceed $\pi/4$.

We reserve $S_1, S_2, \dots$, to denote either squares or S_0 . A side of $S(\alpha)$ or of a square always includes its endpoints.

Let

$$c = 10^{-10}. \quad (3)$$

A *good* square is one whose inclination is at most c and a *bad* square is one that is not good. In a good square two sides are nearly horizontal. Similarly we use the terms nearly vertical, and right, left, top and bottom. We remark that we have introduced the concept of a good square because it is of crucial importance in the proof of Lemma 6 that the angle $\phi(X)$ considered there is smaller than $\pi/4$.

We use X and Y to denote typical points on the bottom and left sides respectively of the containing square.

The inclination of a line segment is the least angle that it makes with the horizontal sides of the containing square, and if it is parallel to the horizontal or vertical sides of $S(\alpha)$, then we refer to it as horizontal or vertical respectively.

We use I to denote a typical interval (open, half open, or closed) of points on a line segment, and we reserve J for finite unions of such intervals (not necessarily on the same line segment) with mutually disjoint interiors. The length of J is the sum of the lengths of the constituent intervals and is denoted by $le J$. We reserve E for finite unions of disjoint intervals of points Y on the left side of $S(\alpha)$.

The waste region $W_{\mathcal{A}}$ in a packing $\mathcal{A}$ is the set of points of $S(\alpha)$ which are neither in any member of $\mathcal{A}$ nor on the boundary of $S(\alpha)$. Let $w_{\mathcal{A}}(\alpha)$ denote the area of $W_{\mathcal{A}}$. Then, by (1),

$$w(\alpha) = \inf_{\mathcal{A}} w_{\mathcal{A}}(\alpha) \quad (4)$$

where $\mathcal{A}$ ranges over all packings of $S(\alpha)$.

Given two points P, Q we use $d(P, Q)$ to denote the distance between P and Q . For two given sets A, B of points we define the vertical distance $f(A, B)$ (relative to $S(\alpha)$) between A and B as follows. Let L be any vertical line segment with endpoints $P = P(L)$ and $Q = Q(L)$ in A and B , respectively. If no such L exists, then take $f(A, B) = \infty$. Otherwise write $f(A, B) = \inf d(P, Q)$ where the infimum is taken over all such L .

3. AN IMPORTANT LEMMA

At many critical points in the proof of the theorem we appeal to the following fundamental lemma. During the proof of both this and the theorem we make use of the observation that, given $S_1, \dots, S_i$, the inclination between S_1 and S_i is at most $\sum_{j=1}^{i-1} \theta_j$ where θ_j is the inclination between S_j and S_{j+1} . This is easily shown by induction.

FUNDAMENTAL LEMMA. *Let $\mathcal{A}$ be a packing, suppose that $S_1 \in \mathcal{A}$ and suppose that either $S_2 \in \mathcal{A}$ or $S_2 = S_0$. Let θ be the inclination between S_1 and S_2 , and suppose that the distance from S_1 to S_2 is at most 1. Then there is an open disc of radius 2 containing both S_1 and a subset of $W_{\mathcal{A}}$ having area at least $c\theta$.*

Proof. We can certainly suppose that $\theta > 0$. We construct a closed rectangle R as follows. When $S_2 \in \mathcal{A}$ let L be the line segment joining the centres of S_1 and S_2 , and when $S_2 = S_0$ let L be any shortest line segment joining the centre of S_1 to S_0 . The length of L is at most $1 + \sqrt{2}$. Consider the two line segments perpendicular to L , each of length 1 and each having their midpoints at the endpoints of L . The endpoints of these two line segments are the four vertices of R .

The number m of squares in $\mathcal{A}$ (including S_1 , and S_2 when $S_2 \neq S_0$) that can intersect R satisfies $m \leq 20$ since every such square is completely contained in a rectangle of area at most $(1 + 2\sqrt{2})(1 + 3\sqrt{2}) < 21$. We divide R into $4m + 1$ equal strips of width $1/(4m + 1)$ and running parallel to L . Each strip connects S_1 to S_2 , and there is at least one that contains no vertex, in its interior, of any square. Such a strip is divided into trapezoidal regions by those squares which have sides crossing the strip. Since $\theta > 0$

at least one of these trapezoidal regions is such that its interior is in $W_{\mathcal{A}}$, its parallel sides are a distance $1/(4m+1)$ apart and its other two sides are inclined at an angle of at least θ/m to one another. Thus its area is at least

$$(4m+1)^{-2} \tan \frac{\theta}{2m} > c\theta$$

and is contained, together with S_1 , in the open disc centre the midpoint of L and radius 2.

4. PROOF OF THE THEOREM

Let

$$c_0 = c^{10} \tag{5}$$

and suppose that there is a packing $\mathcal{A}$ of $S(\alpha)$ such that

$$w_{\mathcal{A}}(\alpha) < c_0(\|\alpha\| \alpha)^{1/2}. \tag{6}$$

Corresponding to each point Y , not a vertex of $S(\alpha)$, on the left side of $S(\alpha)$ we associate a *path* $\pi(Y)$ consisting of a set of line segments $L_1, L_2, \dots$, joined end to end and constructed in the following way. Consider the horizontal line L' through Y . If Y lies on the boundary of a member of $\mathcal{A}$, then let $P_1 = Y$. Otherwise let P_1 be the first point, strictly to the right of Y , on L' that is not in $W_{\mathcal{A}}$. Let L_1 be the line segment joining Y to P_1 . We now proceed inductively for $k = 1, 2, \dots$, as follows. If P_k lies on a bad square, a nearly horizontal side of a good square, or S_0 , then the path $\pi(Y)$ is terminated at P_k and consists of the line segments $L_1, L_2, \dots, L_k$. If the remaining possibility occurs, namely that P_k lies on a nearly vertical side of, but is not a vertex of, a good square S_k , then consider the line through P_k and parallel to the nearly horizontal sides of S_k . Let P_{k+1} be the first point on this line to the right of P_k that is in $S_0 \cup (\bigcup_{S \in \mathcal{A} \text{ } S \neq S_k} S)$. Let L_{k+1} be the line segment joining P_k to P_{k+1} .

Clearly, after a finite number of steps the path will terminate. Whenever we refer to a line segment of a path $\pi(Y)$ we mean one of the line segments $L_1, L_2, \dots$, that occur in the above construction. Paths that terminate on the right side of $S(\alpha)$ are called *chains*, and the line segments of chains are called *links*.

We make several observations regarding situations which occur, sometimes in somewhat different forms, on a number of occasions in the proof. First of all, there are at most 50 squares contained entirely in a given disc of radius 4, so that if we invoke the fundamental lemma for each of a collection of pairs S_1 and S_2 , then no point of $W_{\mathcal{A}}$ can be in more than 2500 of

the subregions of $W_{\mathcal{A}}$ that are provided by the lemma. Second, when passing along paths there are two cases, namely (a) passing from S_0 to a member of $\mathcal{A}$ and (b) passing from one member of $\mathcal{A}$ to another. The same method of proof usually applies to both cases, as we have already seen in the fundamental lemma, and is the cause of the artificial form of some statements.

Third, there is a principle that is used without comment in Lemmas 4 and 6. Let R be a rectangle whose sides are parallel to the sides of $S(\alpha)$ and which has one side on each of the top and bottom sides of $S(\alpha)$. Label the sides of R as left, right, top, and bottom in the obvious way and let I be the bottom side of R . Let $h_i(X)$ ($i = 1, 2$) be piecewise linear functions of the distance of X from the left endpoint of I , and such that $0 \leq h_i(X) \leq \alpha$ for $X \in I$. Consider the sets A_i ($i = 1, 2$) of points P such that if X is the projection of P onto I , then P is at a distance $h_i(X)$ from I . For example, the A_i could be horizontal lines, or portions of paths that cross R . A point is between A_1 and A_2 if it lies on a vertical line segment which has its endpoints on A_1 and A_2 . Let B be the set of points that lie between A_1 and A_2 , and for a given $X \in I$ let $L(X)$ denote that vertical line segment with endpoints on A_1 and A_2 whose projection on I is X . The principle that we use is simply that

$$\text{Area}(B \cap W_{\mathcal{A}}) = \int_I \text{le}(L(X) \cap W_{\mathcal{A}}) dX.$$

One way of proving this heuristically obvious relation is to observe that I can be dissected into a finite union of disjoint intervals by those X for which either $L(X)$ passes through a vertex of a square or for which a derivative $h_i'(X)$ ($i = 1, 2$) does not exist. On each of these subintervals I^* , $\text{le}(L(X) \cap W_{\mathcal{A}})$ is linear and the corresponding subregion of $B \cap W_{\mathcal{A}}$ consists of a finite number of trapeziums with vertical parallel sides a distance $\text{le } I^*$ apart. It is easily seen that their area is

$$\int_{I^*} (L(X) \cap W_{\mathcal{A}}) dX$$

and summing over the I^* gives the desired conclusion.

Before proceeding with the proof of the theorem we give an outline of our procedure. The main part of the proof is to establish, using (6), that there are r chains $\pi(Y_1), \dots, \pi(Y_r)$, where $r > c\alpha$, such that

- (i) $\pi(Y_i)$ terminates on the right side of $S(\alpha)$,
- (ii) $\text{le}(\pi(Y_i) \cap W_{\mathcal{A}}) \leq c \|\alpha\|$,
- (iii) $f(\pi(Y_i), \pi(Y_j)) \geq 20$ ($i \neq j$).

The properties (i) and (ii) imply that $\pi(Y_i)$ passes through at least $1 + [\alpha]$ good squares. An examination of the projection onto the bottom side of $S(\alpha)$ of that part of $\pi(Y_i)$ not in $W_{\mathcal{A}}$ reveals that at least one of the squares

through which $\pi(Y_i)$ passes has inclination at least $c(\|\alpha\|/\alpha)^{1/2}$. Thus, by applying the fundamental lemma successively as we travel from left to right along $\pi(Y_i)$ it follows that there is a subset of $W_{\mathcal{A}}$, every point of which is at most a distance 4 from $\pi(Y_i)$, having area at least $c^3(\|\alpha\|/\alpha)^{1/2}$. Summing over the different $\pi(Y_i)$ and using (iii) gives a contradiction to (6).

We say that two paths merge if at least one of the line segments of one is common to the other. Clearly paths can merge only at nearly vertical sides of good squares. Since there are only a finite number of squares and each square can only focus at most one path onto a given point, it follows that there are only a finite number of paths which merge with a given path and only a finite number of paths which meet vertices of squares.

DEFINITION. Let E be the set of those points Y (not vertices of S_0) for which $\pi(Y)$ does not meet a vertex of a square.

Then E is a finite union of disjoint open intervals and

$$\text{le } E = \alpha. \quad (7)$$

DEFINITION. Given an interval I of points Y we say that it has property (Ω) if the following is true. Every path $\pi(Y)$ with $Y \in I$ passes through the same good squares in the same order and terminates on the same line segment.

There are two important consequences of (Ω) . First of all, if $Y \in I$, $Y' \in I$ and L_j, L'_j are the j th line segments of $\pi(Y), \pi(Y')$ respectively, then L_j and L'_j are parallel and are a distance of at least $d(Y, Y')$ apart.

Second, consider a line segment L which meets each of the paths $\pi(Y)$, with $Y \in I$, on their j th line segments and which meets none of these paths elsewhere. The set of points on L which also lie on paths $\pi(Y)$ with $Y \in I$ form an interval I_L . For each subinterval I_L^* of I_L there corresponds a subinterval I^* of I such that every path $\pi(Y)$ with $Y \in I$ that meets I_L^* has $Y \in I^*$, such that every path $\pi(Y)$ with $Y \in I^*$ meets I_L^* , and such that for every $P \in I_L^*$ there is a unique Y with $Y \in I^*$ for which P lies on $\pi(Y)$. In fact, the paths π induce a linear mapping from I to I_L .

LEMMA 1. *Every interval in E has property (Ω) .*

We leave the proof of this to the reader.

We now modify some of the paths. However, all those that are modified will be later discarded. We write E as a finite union of intervals and let I denote a typical member of this union. Let $Y \in I$ and let Y_0 denote the lower endpoint of I . Clearly $\pi(Y) \cap W_{\mathcal{A}}$ is a finite union (of type J) of open intervals. Let $g(Y) = \text{le}(\pi(Y) \cap W_{\mathcal{A}})$. Then, since I possesses property (Ω) , $g(Y)$ is a linear function of $d(Y, Y_0)$, depending only on the configuration of the squares through which the paths $\pi(Y)$ pass and the line segment on

which they terminate. Thus I is the union of two disjoint intervals (one may possibly be empty) I_0' and I_0'' such that for $Y \in I_0'$ we have $g(Y) < c \|\alpha\|$ and for $Y \in I_0''$ we have $g(Y) \geq c \|\alpha\|$. Let $L_1(Y), L_2(Y), \dots$, be the line segments of $\pi(Y)$ and let

$$g_j(Y) = \sum_{i=1}^j \text{le}(L_i \cap W_{\mathcal{A}}).$$

We now define intervals $I_1', I_1'', I_2', I_2'', \dots$, inductively as follows. The function $g_j(Y)$ is a linear function of the distance of Y from the lower endpoint of I_{j-1}' , and so I_{j-1}'' is the union of two disjoint intervals I_j' and I_j'' (possibly empty) such that for $Y \in I_j'$ we have $g_j(Y) \geq c \|\alpha\|$ and for $Y \in I_j''$ we have $g_j(Y) < c \|\alpha\|$. Thus, if k is the number of line segments of the paths $\pi(Y)$ with $Y \in I$, then I_k'' is empty and I is the union of the disjoint intervals $I_0', I_1', \dots, I_k'$. We obtain modified paths $\pi^*(Y)$ as follows. For $Y \in I_0'$ we take $\pi^*(Y) = \pi(Y)$. For $Y \in I_j'$ ($j = 1, \dots, k$) let $P_j(Y)$ be the left endpoint of $L_j(Y)$ and choose Q on L_j so that if L_j^* is the line segment from P_j to Q , then

$$\text{le}(L_j^* \cap W_{\mathcal{A}}) + \sum_{i=1}^{j-1} \text{le}(L_i \cap W_{\mathcal{A}}) = c \|\alpha\|.$$

We then take $\pi^*(Y)$ to consist of the line segments $L_1, \dots, L_{j-1}, L_j^*$.

By a line segment of $\pi^*(Y)$ we mean one of the line segments $L_1, \dots, L_{j-1}, L_j^*$ arising in the construction of $\pi^*(Y)$. We say that an interval I of points Y has property (Ω^*) when it satisfies (Ω) with π replaced by π^* . In view of Lemma 1 and the remarks above we see that E can be written as a finite union

$$E = \bigcup_i I_i \quad (8)$$

of disjoint intervals I_i each having property (Ω^*) . Clearly the two important consequences of (Ω) mentioned after the definition of (Ω) also hold for (Ω^*) (provided that we replace π by π^*).

We now further modify the paths. Let E' be the set of those $Y \in E$ for which $\pi^*(Y)$ merges with at least one other path. Let S_1, S_2 be given good squares or S_0 , with $S_1 \neq S_2$, and let S_3 be a given good square. Let $J_i = J_i(S_i, S_3)$ ($i = 1, 2$) be the set of those $Y \in E$ such that there are consecutive line segments L_j, L_{j+1} of $\pi^*(Y)$ for which L_j has its initial point on the left side of S_i and L_{j+1} has its initial point on the left side of S_3 . Then J_i is a finite union of disjoint intervals each having property (Ω^*) . Let $J_i' = J_i'(S_i, S_3)$ be the set of those points on the left side of S_3 that lie on a $\pi^*(Y)$ with $Y \in J_i$. Then J_i' is a finite union of disjoint intervals. Consider $J_1' \cap J_2'$. This is also a finite union of disjoint intervals, and hence so is $J_1' \setminus (J_1' \cap J_2')$. Therefore,

by the second important consequence of (Ω^*) , the set J_3 of those $Y \in J_1$ for which $\pi^*(Y)$ has a point in common with $J_1' \cap J_2'$ is a finite union of disjoint intervals, each having property (Ω^*) . The same conclusion holds for $J_4 = J_1 \setminus J_3$. We observe that the $\pi^*(Y)$ with $Y \in J_3$ merge on the left side of S_3 with paths $\pi^*(Y)$ whose previous line segment emanates from S_2 , whereas those with $Y \in J_4$ do not. Clearly $E' = \bigcup J_3(S_1, S_2, S_3)$ where the union is taken over all triples S_1, S_2, S_3 . Thus $E \setminus E' = \bigcap J_4(S_1, S_2, S_3)$, and both E' and $E \setminus E'$ can be written as finite unions

$$E' = \bigcup_i I_i^*, \quad E \setminus E' = \bigcup_i I_i^{**} \quad (9)$$

of disjoint intervals, chosen so that they all have property (Ω^*) and so that each interval I_i^* has the following further properties.

(a) For every $Y \in I_i^*$ the $\pi^*(Y)$ merge with other paths on the left sides of exactly the same good squares.

(b) If S_1 is a good square on which the paths $\pi^*(Y)$ with $Y \in I_i^*$ merge with other paths, then there is an S_2 (good or S_0) such that the $\pi^*(Y)$ with $Y \in I_i^*$ merge on S_1 with paths $\pi^*(Y')$ that have just emanated from S_2 (of course, they may merge with other paths also).

For $Y \in E \setminus E'$ we take $\pi^{**}(Y) = \pi^*(Y)$. For a given $Y \in I_i^{**}$, let $L_1, L_2, \dots$, be the line segments of $\pi^*(Y)$. Let j be the smallest value of i for which L_{i+1} is a line segment of some $\pi^*(Y')$ with $Y' \neq Y$. Let $\pi^{**}(Y)$ consist of the line segments $L_1, L_2, \dots, L_j$, which we refer to as the line segments of $\pi^{**}(Y)$. We say that an interval I of points Y has property (Ω^{**}) if it satisfies (Ω) with π replaced by π^{**} . Clearly the intervals I_i^* and I_i^{**} have property (Ω^{**}) .

We observe that each interval I_i^{**} is of at least one of the following four types, which we refer to as type 1, ..., type 4, respectively.

- (1) For every $Y \in I_i^{**}$ we have $\text{le}(\pi^{**}(Y) \cap W_{\mathcal{A}}) = c \|\alpha\|$.
- (2) For every $Y \in I_i^{**}$, $\pi^{**}(Y)$ terminates on a bad square.
- (3) For every $Y \in I_i^{**}$, $\pi^{**}(Y)$ terminates on a nearly horizontal side of a good square or a horizontal side of $S(\alpha)$.
- (4) For every $Y \in I_i^{**}$, $\pi^{**}(Y)$ terminates on the right side of $S(\alpha)$.

Note that, for every $Y \in E$, $\text{le}(\pi^{**}(Y) \cap W_{\mathcal{A}}) \leq c \|\alpha\|$, and none of the paths $\pi^{**}(Y)$ merge.

Let E_k denote the union of those I_i^{**} which are of type k . Thus, by (7) and (9),

$$\alpha = \text{le } E \leq \text{le } E' + \sum_{k=1}^4 \text{le } E_k. \quad (10)$$

Note that for $Y \in E_4$, the path $\pi^{**}(Y)$ is a chain for which $\pi^{**}(Y) = \pi(Y)$, and (i) and (ii) on page 174 hold with Y_i replaced by Y .

LEMMA 2. *We have $\text{le } E' < c\alpha$.*

Proof. Let S_1, S_2 be given good squares or S_0 , with $S_1 \neq S_2$, and let S_3 be a given good square. Let $J = J(S_1, S_2, S_3)$ be the union of those intervals I_i^* such that if $Y \in I_i^*$, then

- (i) the last line segment of $\pi^{**}(Y)$ has its initial point on the left side of S_1 and final point (the terminal point of $\pi^{**}(Y)$) on the left side of S_3 ,
- (ii) there is a line segment, of a path $\pi^*(Y')$ that merges with $\pi^*(Y)$, which has its initial point on the left side of S_2 and its final point the terminal point of $\pi^{**}(Y)$.

Let $Y_1 \in J$, $Y_2 \in J$, let L_i be the line segment of $\pi^{**}(Y_i)$ whose initial point is on the left side of S_1 and let P_i be the terminal point of $\pi^{**}(Y_i)$. Let L_i^* be the perpendicular from P_i to the left side of S_2 . This is a line segment of a path $\pi^*(Y)$ and hence S_1 and S_2 are a distance of at most $2c \|\alpha\| \leq c$ apart. For either $i = 1$ or $i = 2$ the line segment L_i^* intersects with L_{3-i} , and if necessary by relabeling, we can suppose that $i = 1$. Let P be the point of intersection of L_1^* and L_2 . Let L_3 be the perpendicular from P to L_1 and let Q be the point of intersection of L_3 and L_1 . Since L_1^* and L_2 intersect at P and are both line segments of paths $\pi^*(Y)$, the point P , and hence also the point Q , must be in $W_{\mathcal{A}}$. Let $h = h(Y_1, Y_2)$ be the distance between L_1 and L_2 , i.e., $h = \text{le } L_3$, and let θ be the inclination between S_1 and S_2 . Then

$$h = (\tan \theta) d(Q, P_1) \leq c \|\alpha\| \tan \theta.$$

Since S_1 and S_2 are good squares or S_0 we have $0 \leq \theta \leq 2c$ so that

$$h \leq c\theta. \quad (11)$$

Further, since the paths π^{**} do not merge, and each of the intervals I_i^* has property (Ω^{**}) , it follows that

$$\text{le } J \leq \sup h(Y_1, Y_2)$$

where the supremum is taken over all pairs $Y_1, Y_2 \in J$. Hence, by (11),

$$\text{le } J \leq c\theta. \quad (12)$$

By the fundamental lemma, there is a disc of radius 2 that contains at least

one of S_1 or S_2 together with a subset $V = V(S_1, S_2, S_3)$ of $W_{\mathcal{A}}$ having area at least $c\theta$. Since every disc of radius 10 contains at most 400 members of $\mathcal{A}$, no point of $W_{\mathcal{A}}$ can be the centre of a disc of radius 10 that contains more than 64,000,000 triples S_1, S_2, S_3 of good squares or (in the events $S_1 = S_0$ or $S_2 = S_0$) more than 160,000 pairs S_1, S_3 of good squares or more than 160,000 pairs S_2, S_3 of good squares. Hence no point of $W_{\mathcal{A}}$ can be in more than c^{-1} sets $V(S_1, S_2, S_3)$. Thus, by (12), on summing over all possible triples S_1, S_2, S_3 we obtain

$$\text{le } E' \leq \sum \text{le } J(S_1, S_2, S_3) \leq c^{-1} w_{\mathcal{A}}(\alpha).$$

The lemma now follows at once from (5) and (6).

LEMMA 3. *We have $\text{le } E_1 < c\alpha$.*

Proof. Consider an interval I_i^{**} of type 1. Let $Y_1 \in I_i^{**}$, $Y_2 \in I_i^{**}$. Since I_i^{**} has property (Ω^{**}) the region between $\pi^{**}(Y_1)$ and $\pi^{**}(Y_2)$ is divided up into rectangles, corresponding to interiors of good squares, and trapeziums, whose interiors are in $W_{\mathcal{A}}$. The line segments of $\pi^{**}(Y_1)$ and $\pi^{**}(Y_2)$ are spaced at least $d(Y_1, Y_2)$ apart, and $\text{le}(\pi^{**}(Y_i) \cap W_{\mathcal{A}}) = c \|\alpha\|$. Hence the sum of the areas of the trapeziums is at least $d(Y_1, Y_2) c \|\alpha\|$. Now, on taking the limit as Y_1, Y_2 tend to the lower and upper endpoints of I_i^{**} , respectively, we see that associated with I_i^{**} there is an open subset V_i of $W_{\mathcal{A}}$ which has area at least $c \|\alpha\| \text{le } I_i^{**}$.

For every point $P \in V_i$ there is a $Y \in I_i^{**}$ such that $\pi^{**}(Y)$ passes through P . Since paths $\pi^{**}(Y')$ do not merge a good square can focus a line segment of at most one path $\pi^{**}(Y')$ with $Y' \in E$ onto P . Thus P can be in at most $\pi(\sqrt{2} + c \|\alpha\|)^2 < 9$ different V_i . Hence, on summing over the I_i^{**} of type 1 we obtain

$$c \|\alpha\| \text{le } E_1 \leq 9 w_{\mathcal{A}}(\alpha).$$

This with (2), (5), and (6) gives Lemma 3.

LEMMA 4. *We have $\text{le } E_2 < c\alpha$.*

Proof. It suffices to show that the number of bad squares is at most $c^2\alpha$. For no point on the side of a bad square can be on more than 9 different paths $\pi^{**}(Y)$ and so, because of property (Ω^{**}) , the sum of the lengths of those intervals of type 2 can be at most 9 times the sum of the lengths of the sides of the bad squares. This gives the desired conclusion.

We now proceed to show that the number of bad squares is at most $c^2\alpha$. Let J, J' denote the projections of the bad squares onto the bottom and left sides respectively of the containing square. Then J and J' are the unions of a finite number of disjoint closed intervals. Let a and b denote the lengths of

J and J' , respectively. Then the total number of bad squares is at most ab since they are contained in a finite number of rectangles whose total area is ab . It therefore suffices to show that

$$a < c\alpha^{1/2}, \quad b < c\alpha^{1/2}. \quad (13)$$

Since the arguments are identical we only treat J . We can certainly suppose that $a > 0$. Consider the projection of a single bad square onto the bottom side of the containing square. This is a closed interval I of length h , say, where $1 \leq h \leq \sqrt{2}$. For each $X \in I$ let $L(X)$ denote the vertical line from X to the top side of $S(\alpha)$. We observe that the points of $L(X) \cap W_{\mathcal{A}}$ are a finite union of open intervals. Let $g(X) = \text{le}(L(X) \cap W_{\mathcal{A}})$ and let B be the set of those $X \in I$ for which $g(X) \geq c$. There are two cases, $B = I$ and $B \neq I$.

If $B = I$, then the waste space vertically above I has area

$$\int_I g(X) dX \geq ch. \quad (14)$$

If, on the other hand, $B \neq I$, then there is an $X \in I$ such that $g(X) < c$. Let $S_1, \dots, S_k$ be the squares that we meet, in order, on traveling up $L(X)$ (with an obvious convention when we meet two or more simultaneously). Let θ_1 be the inclination of S_1 and θ_j ($j = 2, \dots, k$) the inclination between S_{j-1} and S_j . Then S_1 is at most a distance c from S_0 , and S_j is at most a distance c from S_{j-1} ($j = 2, \dots, k$). Hence, by the fundamental lemma there is a disc of radius 2 about S_j which contains a subset V_j of $W_{\mathcal{A}}$ having area at least $c\theta_j$. We sum over the V_j . Hence the total area of the set of those points of $W_{\mathcal{A}}$ that are within a distance 4 from $L(X)$ is at least

$$\frac{1}{50} \sum_{j=1}^k c\theta_j.$$

The line $L(X)$ meets a bad square. Hence $\sum_j \theta_j > c$. Therefore, by (3) and (14), in either case there is an interval of length 8 on the bottom side of $S(\alpha)$ which contains I and vertically above which the area of waste space is at least c^3 .

Since $a > 0$, we can always choose $\max(1, [a/20]) \geq a/40$ bad squares whose projections onto the bottom side of $S(\alpha)$ are spaced a distance apart of at least 16. Hence the area of $W_{\mathcal{A}}$ is at least $ac^3/40$. This with (3), (5), and (6) gives (13) and hence completes the proof of the lemma.

LEMMA 5. *We have $\text{le } E_3 < c\alpha$.*

We leave the finer details of this to the reader, as the proof is similar to that of Lemma 2. Let S_1 (good or S_0) and S_2 (good or S_0) be given and consider the union J of those intervals I_i^{**} such that if $Y \in I_i^{**}$, then the last line

segment of $\pi^{**}(Y)$ has its initial point on the left side of S_1 and final point on a nearly horizontal side of S_2 . This nearly horizontal side of S_2 plays the same rôle as the line segment L_1^* occurring in the proof of Lemma 2.

Let $Y_1 \in J$, $Y_2 \in J$ and let $h = h(Y_1, Y_2)$ be the distance between the last line segments of $\pi^{**}(Y_1)$ and $\pi^{**}(Y_2)$. As in the proof of Lemma 2, we obtain

$$h \leq c \|\alpha\| \tan \theta \leq c\theta,$$

where θ is the inclination between S_1 and S_2 . Since the paths π^{**} do not merge, and each interval I_i^{**} has property (Ω^{**}) , it follows that

$$\text{le } J \leq \sup h(Y_1, Y_2) \leq c\theta,$$

where the supremum is taken over all pairs $Y_1, Y_2 \in J$. On appealing to the fundamental lemma and summing over all pairs S_1, S_2 we conclude that

$$\text{le } E_3 < c^{-1} w_{\mathcal{A}}(\alpha)$$

and this with (5) and (6) gives Lemma 5.

We can now conclude from (3), (10) and Lemmas 2, 3, 4, and 5, that

$$\text{le } E_4 > \frac{3}{4}\alpha. \quad (15)$$

We wish to choose $\gg \alpha$ different Y in E_4 , say $Y_1, Y_2, \dots$, such that, if possible, (iii) on page 174 holds. We proceed by choosing $\gg \alpha$ paths $\pi(Y)$ whose initial points Y are spaced at least a distance 40 apart. We will then show that only a small proportion of the paths chosen can come appreciably closer together.

Suppose, for the time being, that $\alpha \geq 160$, and let I_i ($i \leq \frac{1}{2} + \alpha/80$) denote the open interval on the left side of S_0 whose endpoints are at distances of $80i - 80$ and $80i - 40$ from the bottom left vertex. Let M denote the number of different i for which $I_i \cap E_4$ is empty. Then, by (15),

$$M < \alpha/160.$$

Hence there are at least

$$\left\lceil \frac{1}{2} + \frac{\alpha}{80} \right\rceil - \left\lceil \frac{\alpha}{160} \right\rceil \geq \left\lceil \frac{\alpha}{160} \right\rceil > \frac{\alpha}{320}$$

different i for which $I_i \cap E_4$ is not empty. We choose one point Y from each set $I_i \cap E_4$. Thus we have q chains $\pi(Q_1), \dots, \pi(Q_a)$, where

$$q > \alpha/320, \quad (16)$$

such that Q_{i+1} is above Q_i , $d(Q_{i+1}, Q_i) > 40$, and each chain $\pi(Q_i)$ satisfies (i) and (ii) on page 174 (with $Y_i = Q_i$).

If, instead, $\alpha < 160$, then we have the same conclusion on simply taking $q = 1$ and choosing Q_1 to be any element of E_4 .

For two given sets of points we define the *variation* between them as

$$\sup_{L, L'} (le L - le L')$$

where the supremum is taken over all pairs of vertical line segments L, L' joining the sets. We shall not consider the variation between pairs of sets for which no such line segments exist. We say that a point is between $\pi(Q_i)$ and $\pi(Q_j)$ if it lies on a vertical line segment with endpoints on $\pi(Q_i)$ and $\pi(Q_j)$.

LEMMA 6. *Let R be a rectangle with two opposite sides on the top and bottom sides of $S(\alpha)$, and let $v_R(i, j)$ denote the variation between $\pi(Q_i) \cap R$ and $\pi(Q_j) \cap R$. Further, let $G_R(i, j)$ denote the set of those points of R that lie between $\pi(Q_i)$ and $\pi(Q_j)$, and let $H_R(i, j)$ denote the set of those points that are within a distance of 5 from $G_R(i, j)$. Then $H_R(i, j) \cap W_{\mathcal{A}}$ has area at least $c^3 v_R(i, j)$.*

Proof. Let h denote the length of the bottom side of R . Divide R into $n = [h] + 1$ equal closed vertical strips, $T_1, \dots, T_n$, each of width $h/([h] + 1)$, and let I_k denote the bottom side of T_k . Let $T_k^* = T_k \cap G_R(i, j)$, and let v_k denote the variation between $\pi(Y_i) \cap T_k$ and $\pi(Y_j) \cap T_k$. Since the links of $\pi(Y_i)$ and $\pi(Y_j)$ are inclined at angles of at most c to the horizontal it follows that

$$v_k \leq 2 \tan c < 3c.$$

Also, we have

$$\sum_{k=1}^n v_k \geq v_R(i, j). \quad (17)$$

For each point $X \in I_k$ let $L(X)$ denote the vertical line segment with endpoints on $\pi(Q_i)$ and $\pi(Q_j)$ whose projection on I_k is X . Let $\phi(X)$ be the least angle between the links of $\pi(Q_i)$ and $\pi(Q_j)$ that meet $L(X)$ from the left. Clearly $\phi(X)$ is piecewise constant.

We first show that the set F_k of those points of $W_{\mathcal{A}}$ within a distance 4 of T_k^* satisfies

$$\text{area } F_k \geq c^2 v_k. \quad (18)$$

We can certainly suppose that $v_k > 0$. Let J denote the set of those points $X \in I_k$ for which $\phi(X) > v_k/4$. Since ϕ is piecewise constant J is a finite union of disjoint intervals, and hence so is $I_k \setminus J$. Then the sum of the variations

between $\pi(Q_i)$ and $\pi(Q_j)$ on the intervals making up $I_k \setminus J$ can be at most

$$(1 - \text{le } J)(\tan c - \tan(c - \frac{1}{4}v_k)) < \frac{1}{2}v_k.$$

Hence the sum of the variations between $\pi(Q_i)$ and $\pi(Q_j)$ on the intervals making up J is at least $v_k/2$. Therefore, we have $2(\text{le } J) \tan c \geq v_k/2$, so that

$$\text{le } J \geq v_k/8c. \quad (19)$$

If for every $X \in J$ we have $\text{le}(L(X) \cap W_{\mathcal{A}}) \geq c$, then

$$\text{area}(T_k^* \cap W_{\mathcal{A}}) \geq \int_J \text{le}(L(X) \cap W_{\mathcal{A}}) dX \geq c \text{le } J.$$

The inequality (18) then follows from (19). Thus we can suppose that there is an $X \in J$ such that

$$\text{le}(L(X) \cap W_{\mathcal{A}}) < c. \quad (20)$$

Let S_1 be the square on the link of $\pi(Q_i)$ that meets $L(X)$ from the left (if this link is the first one, then let $S_1 = S_0$). Let $S_2, \dots, S_{t-1}$ be the squares, other than S_1, S_t , that $L(X)$ meets as we travel along $L(X)$ from $\pi(Q_i)$ to $\pi(Q_j)$, and let S_t be the square on the link of $\pi(Q_j)$ that meets $L(X)$ from the left, with the same convention as for S_1 . Let θ_r be the inclination between S_r and S_{r+1} ($r = 1, \dots, t-1$). As we have already observed in Section 3, $\sum_{r=1}^{t-1} \theta_r$ is at least as large as the inclination between S_1 and S_t . Further, since S_1 and S_t are good or S_0 , so that $0 \leq \phi(X) \leq 2c < \pi/4$, the inclination between S_1 and S_t is equal to $\phi(X)$. Therefore, since $X \in J$,

$$\sum_{r=1}^{t-1} \theta_r > \frac{1}{4}v_k. \quad (21)$$

By (20) and the fact that $\text{le}(\pi(Q_i) \cap W_{\mathcal{A}}) \leq c \|\alpha\|$ (and similarly for j), the distance between S_r and S_{r+1} is at most $2c$. Hence, by the fundamental lemma, there is an open disc of radius 2 containing either S_r or S_{r+1} and a subset of $W_{\mathcal{A}}$ having area at least $c\theta_r$. On summing over the S_r and using a by now familiar argument we obtain

$$\text{area } F_k > \frac{c}{100} \sum_{r=1}^{t-1} \theta_r.$$

This with (21) gives (18) once more.

If $h < 1$, then $n = [h] + 1 = 1$ and the lemma is an immediate consequence of (17) and (18). If $h \geq 1$, then the width of each strip T_k , $h/([h] + 1)$, lies in the range $[\frac{1}{2}, 1]$. Hence no point can be in more than 17 different F_k

and each F_k is contained in $H_R(i, j)$. Thus, by (17) and (18) we have the lemma once more.

We now show that most of the chains $\pi(Q_i)$ do not come too close together. Let m_1 be the least m such that $f(\pi(Q_1), \pi(Q_{1+m})) \geq 20m$, let m_2 be the least m such that $f(\pi(Q_{1+m_1}), \pi(Q_{1+m_1+m})) \geq 20m$, and so on. This process stops after, say, t steps where $t \geq 0$. Thus¹

$$f(\pi(Q_{1+\dots+m_{j-1}}), \pi(Q_{m_1+\dots+m_j})) \leq 20(m_j - 1).$$

Now we consider two cases. First, suppose that

$$f(\pi(Q_{1+\dots+m_{j-1}}), \pi(Q_{1+\dots+m_j})) \geq 30(m_j - 1).$$

Since $d(Q_{1+\dots+m_{j-1}}, Q_{m_1+\dots+m_j}) \geq 40(m_j - 1)$ there is a rectangle R_j of the form considered in Lemma 6 and such that

$$v_{R_j}(1 + \dots + m_{j-1}, m_1 + \dots + m_j) \geq 10(m_j - 1)$$

and $\pi(Q_{m_1+\dots+m_j}) \cap R_j$ lies between $\pi(Q_{1+\dots+m_{j-1}}) \cap R_j$ and $\pi(Q_{1+\dots+m_j}) \cap R_j$. Hence,

$$\begin{aligned} \text{area}(H_{R_j}(1 + \dots + m_{j-1}, 1 + \dots + m_j) \cap W_{\mathcal{A}}) \\ \geq \text{area}(H_{R_j}(1 + \dots + m_{j-1}, m_1 + \dots + m_j) \cap W_{\mathcal{A}}) \end{aligned}$$

and, by Lemma 6,

$$\text{area}(H_{R_j}(1 + \dots + m_{j-1}, m_1 + \dots + m_j) \cap W_{\mathcal{A}}) \geq 10c^3(m_j - 1).$$

Therefore,

$$\text{area}(H_{R_j}(1 + \dots + m_{j-1}, 1 + \dots + m_j) \cap W_{\mathcal{A}}) \geq 10c^3(m_j - 1). \quad (22)$$

Second suppose that

$$f(\pi(Q_{1+\dots+m_{j-1}}), \pi(Q_{1+\dots+m_j})) < 30(m_j - 1).$$

Then, on taking $R_j = S(\alpha)$, we have

$$v_{R_j}(1 + \dots + m_{j-1}, 1 + \dots + m_j) > 10(m_j - 1).$$

Hence, by Lemma 6, we have (22) once more.

As links of chains have inclination at most c to the horizontal, $n = q - (1 + m_1 + \dots + m_t)$ satisfies $40n < 20n + 2\alpha \tan c$. Thus $n < c\alpha$.

We now sum (22) over the j . Since the chains $\pi(Q_{1+\dots+m_j})$ satisfy

¹ When $j = 1$ we interpret $1 + \dots + m_{j-1}$ as 1.

$f(\pi(Q_{1+\dots+m_{j-1}}), \pi(Q_{1+\dots+m_j})) \geq 20m_j$ no point can be in more than two different H_{R_j} . Hence

$$10c^3 \sum_j (m_j - 1) \leq 2w_{\mathcal{A}}(\alpha)$$

and so, by (3), (5), (6), and (16)

$$q - n = 1 + \sum_j m_j < 1 + \sum_j 1 + c^6 \alpha,$$

$$t + 1 > c\alpha.$$

Let $Y_j = Q_{1+\dots+m_{j-1}}$ ($j = 1, \dots, t + 1$). Then we have r chains $\pi(Y_1), \dots, \pi(Y_r)$, where $r = t + 1 > c\alpha$, which satisfy (i), (ii) and (iii) on page 174. Since $\text{le}(\pi(Y_j) \cap W_{\mathcal{A}}) \leq c \|\alpha\|$, $\pi(Y_j)$ has at least $[\alpha] + 1$ links excluding the first, say k links. Let S_{i-1} ($i = 2, \dots, k + 1$) be the square on the i th link from the left side of $S(\alpha)$, and let θ_i be the inclination of S_i . Then the projection of $S_i \cap \pi(Y_j)$ on the bottom side of $S(\alpha)$ is an interval of length $\cos \theta_i$, and since these intervals do not overlap we have

$$\sum_{i=1}^k (1 - \cos \theta_i) \geq k - \alpha.$$

Hence

$$\max_i \theta_i^2 \geq 1 - (\alpha/k)$$

so that

$$\max_i \theta_i > c(\|\alpha\|/\alpha)^{1/2}. \quad (23)$$

Let $\phi_1 = \theta_1$, and let ϕ_i ($i = 2, \dots, k$) be the inclination between S_{i-1} and S_i . Then

$$\sum_{i=1}^k \phi_i \geq \max_i \theta_i. \quad (24)$$

By the fundamental lemma there is an open disc of radius 2 which contains S_i and a subset of $W_{\mathcal{A}}$ having area at least $c\phi_i$. Hence, on summing over the i , the area of the set of those points of $W_{\mathcal{A}}$ within a distance 4 of $\pi(Y_j)$ is at least

$$\frac{c}{50} \sum_{i=1}^k \phi_i.$$

By (23) and (24) this is at least $c^3(\|\alpha\|/\alpha)^{1/2}$. We recall that $f(\pi(Y_i), \pi(Y_j)) \geq 20$ ($i \neq j$). Thus, on summing over the r ($> c\alpha$) chains $\pi(Y_i)$ we find that the area of $W_{\mathcal{A}}$ is at least $c^4(\|\alpha\|/\alpha)^{1/2}$. This with (3) and (5) contradicts (6).

Therefore, for every packing $\mathcal{A}$ of $S(\alpha)$, $w_{\mathcal{A}}(\alpha) \geq c_0(\|\alpha\| \alpha)^{1/2}$. This with (4) gives the theorem.

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